

Lab Session Task:

Show positive and negative effects of including the momentum term in the gradient descent algorithm with a constant step size using the following very simple problems and some other test problems.

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \alpha \nabla f(\mathbf{x}^{(k)})$$


$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \alpha \nabla f(\mathbf{x}^{(k)}) + \beta(\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)})$$

(1) Minimize $f(\mathbf{x}) = (x_1 + 1)^2 / 9 + (x_2 + 1)^2$

(2) Minimize $f(\mathbf{x}) = (1 - x_1)^2 + 100(x_2 - x_1^2)^2$

(3) Minimize $f(\mathbf{x}) =$ Your own function for explanations .